

Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models

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Abstract

*Today’s most advanced vision-language models (VLMs) remain proprietary. The strongest open-weight models rely heavily on synthetic data from proprietary VLMs to achieve good performance, effectively distilling these closed VLMs into open ones. As a result, the community has been missing foundational knowledge about how to build performant VLMs from scratch. We present **Molmo**, a new family of VLMs that are state-of-the-art in their class of openness. Our key contribution is a collection of new datasets called **PixMo**, including a dataset of highly detailed image captions for pre-training, a free-form image Q&A dataset for fine-tuning, and an innovative 2D pointing dataset, all collected without the use of external VLMs. The success of our approach relies on careful modeling choices, a well-tuned training pipeline, and, most critically, the quality of our newly collected datasets. Our best-in-class 72B model not only outperforms others in the class of open weight and data models, but also outperforms larger proprietary models including Claude 3.5 Sonnet, and Gemini 1.5 Pro and Flash, second only to GPT-4o based on both academic benchmarks and on a large human evaluation. Our model weights, new datasets, and source code are available at <https://molmo.allenai.org/blog>.*

1. Introduction

Large multimodal models are used ubiquitously today. Proprietary models—GPT-4o, Gemini-1.5 Pro, Claude 3.5

Sonnet—produce comprehensive image descriptions and accurately answer complex visual questions. Unfortunately, the most performant of these vision-language models (VLMs) remain proprietary with neither model weights, data, nor code being publicly released.

To foster scientific exploration, numerous research efforts have attempted to reproduce similar capabilities in open models. Early works, exemplified by LLaVA [69], produced fully open weights and training data but now lag significantly behind the state-of-the-art. More recent, stronger open-weight models have trended towards less open data: the training data may either be proprietary (e.g., [10]) or, in cases where it is released, there is a heavy reliance on *synthetic* data generated by proprietary systems, e.g., models are trained on datasets like ShareGPT4V [15] which uses GPT-4V [88] to generate a large set of detailed image captions. The resulting VLMs, therefore, are effectively *distillations* of proprietary VLMs. As it stands, the scientific community is missing foundational knowledge about how to build performant VLMs *from scratch* (more discussion about this and related work are in the Appendix).

In this work, we present the **Molmo** (Multimodal Open Language Model) family of state-of-the-art open VLMs with released model weights *and* released vision-language training data without any reliance on synthetic data from other VLMs, including proprietary ones. The success of our approach relies on careful model design choices, a well-tuned training pipeline, and most critically, the quality of our new datasets, collectively named **PixMo** (**P**ixels for **M**olmo), which are fully open.

High-quality multimodal data, both for pre-training and

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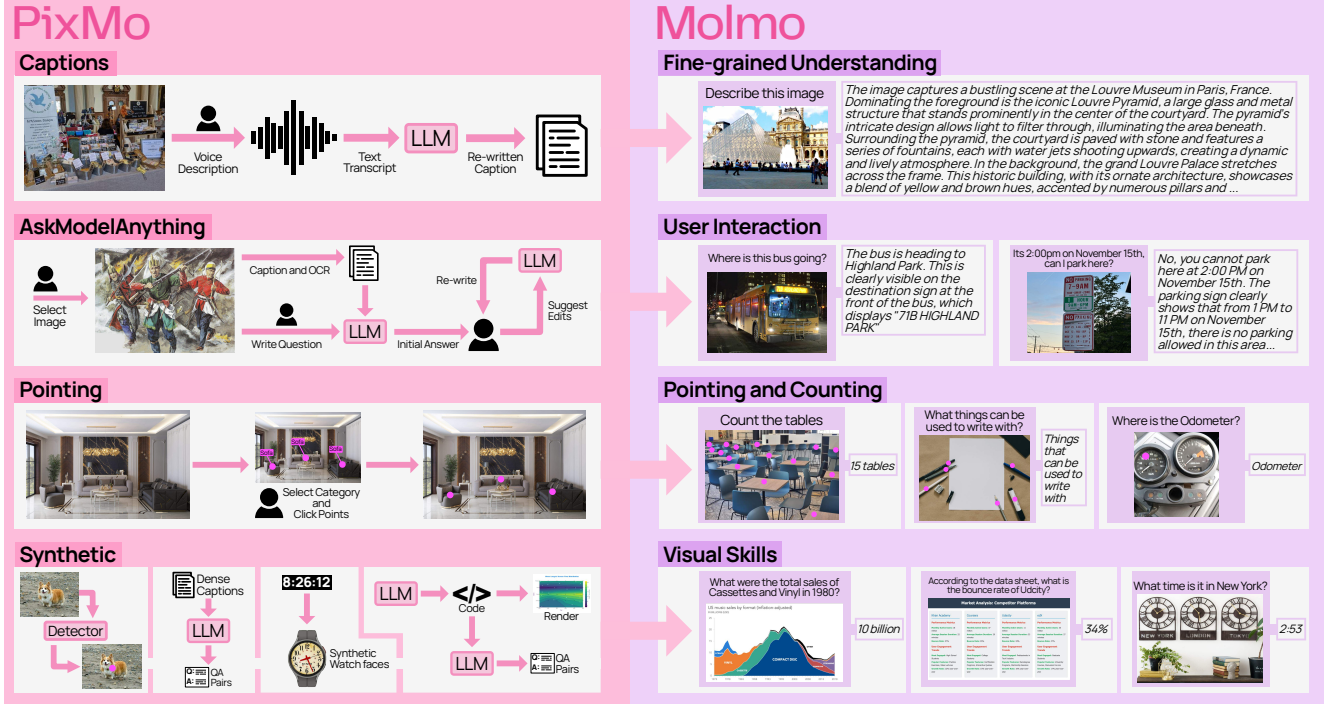


Figure 1. Datasets in **PixMo** (left) and the capabilities they enable in **Molmo** (right). **PixMo** consists of three annotated datasets and four synthetic datasets, all constructed without the use of VLMs. The annotated datasets include: dense captions (for pre-training), instruction following (for fine-tuning), and pointing (for fine-tuning, to support grounding and counting). The four synthetic datasets augment these datasets by targeting additional skills (e.g., clock reading, document understanding).

fine-tuning, is a key missing piece for training open VLMs that are competitive with closed ones. The academic community has struggled to collect such datasets due to high costs and the difficulty of obtaining high-quality annotations. To build **PixMo**, we introduce several data collection innovations that allow us to quickly collect high-quality data from untrained annotators, see Figure 1.

PixMo includes a dataset of 712k images with very long (200+ word) detailed captions. Collecting this data was difficult because directly asking annotators to write such captions produces poor results: they tend to focus on a few salient visual elements [17], typing long paragraphs is time-consuming, and annotators can potentially copy-and-paste responses from proprietary VLMs, circumventing our goal of avoiding distillation. Instead, we ask annotators to describe images in *speech* for 60 to 90 seconds. Empirically, we found that with this modality switching “trick” annotators provide far more detailed descriptions in less time, and for each description we collect an audio receipt (i.e., the annotator’s recording) proving that a VLM was not used.

PixMo also includes an array of fine-tuning datasets. To collect instruction-following data, we have users interactively edit responses with a language-only LLM to obtain high-quality and accurate free-form responses. We gather 162k annotations on 73k images in this way. We also collect a unique new data source that grounds language in images with 2D points. Using points enables us to collect ground-

ing data much faster than would be possible using bounding boxes or segmentation masks since it is much easier to annotate, and we take advantage of this by collecting over 2.3 million grounding annotations for a diverse range of objects, expressions, and scenes. This novel pointing data enables our models to answer some questions more naturally by pointing to the pixels that support the answer, improves counting accuracy (the model *counts by pointing*), and we believe it will open up an important future direction in which VLMs enable agents (e.g., robots, web agents) to *act* by pointing in their environments, e.g., to a navigation waypoint, to an object to pick up, or to a user interface button to press. Finally, we introduce several novel synthetic datasets (meaning with no or minimal human annotations, but still *not* using a VLM) with data targeting particular skills (clock reading, chart understanding, table understanding, etc.) that complements existing open-source datasets.

We train models on these datasets following a mostly standard design using a pre-trained LLM and vision encoder, but with some new improvements including a simplified two-stage training pipeline, a novel overlapping multi-crop strategy, an efficient method of training on images with multiple annotations, and some new insights in how to set up the optimizer and vision/language connector. We evaluate the **Molmo** family of models on 11 academic benchmarks and with a human evaluation that allows us to rank models by user preference. Our most efficient model,

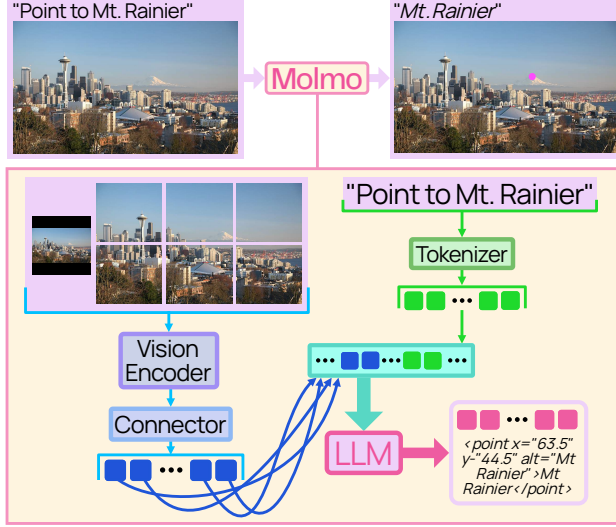


Figure 2. **Molmo** follows the simple and standard design of connecting a vision encoder and a language model.

MolmoE-1B, based on the OLMoE-1B-7B [87] mixture-of-experts LLM, nearly matches the performance of GPT-4V on both our academic benchmarks and user preference. Molmo-7B-O and Molmo-7B-D, based on OLMo-7B [37] and Qwen2 7B [120], respectively, perform comfortably between GPT-4V and GPT-4o [90] on both academic benchmarks and user preference. Our best-in-class Molmo-72B model, based on Qwen2 72B, achieves the highest academic benchmark score and ranks second by human preference, just behind GPT-4o. Our best model outperforms many state-of-the-art proprietary systems, including Gemini 1.5 Pro and Flash [103], and Claude 3.5 Sonnet [7]. We will also release a 100% fully open Molmo model, based on a MetaCLIP [118] vision encoder and OLMo LLM, for which every bit of training data is publicly available. In addition, we perform an expansive set of ablations to better inform the scientific community of how various model and data design choices affect VLMs.

2. Architecture

Our model architecture (Figure 2) follows a standard design, combining pre-trained language and vision models (e.g., [69]). It has four components: (1) a pre-processor that converts the input image into multiscale, multi-crop images, (2) a ViT image encoder [31] that computes per-patch features for each image independently, (3) a connector that pools and projects patch features into the LLM’s embedding space, and (4) a decoder-only LLM [95, 109].

From this template, we build a family of models by selecting a vision encoder and LLM, keeping the training data and recipe consistent across choices (except for learning rates). We primarily use OpenAI’s ViT-L/14 336px CLIP model [96] due to strong performance in initial experiments,

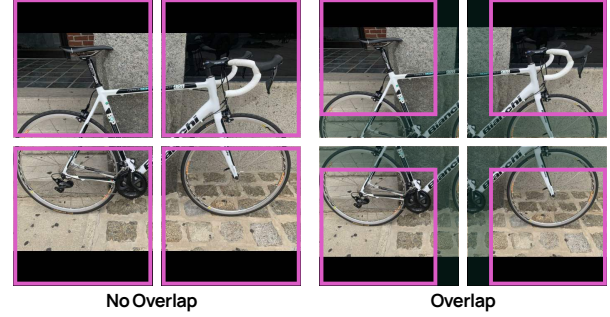


Figure 3. An image cropped without (left) and with (right) overlap. Highlighted regions show areas used by the LLM. Overlapping crops ensure that central patches are encoded with neighboring context; for example, the patches containing the bike’s brand name are always part of a crop where the entire name is visible.

but similar results are achievable with SigLIP [130] and the fully open MetaCLIP [118] (see Section 6). For the LLM, we experiment across scales and openness levels: fully open OLMo-7B-1024-preview, fully open OLMoE-1B-7B (our most efficient model), open-weight Qwen2 7B [120], and open-weight Qwen2 72B (our best-performing model).

Cropping. Most ViTs only support square images at a fixed resolution that is generally too low for fine-grained tasks such as OCR or detailed captioning. To address this issue, we follow recent works [19, 30, 70, 85, 124] by dividing the image into multiple square crops that tile the image. Additionally, the full image, resized to the ViT’s resolution, provides a low-resolution overview. Each crop is processed independently by the ViT. See the Appendix for details.

One limitation of cropping is that border patches lack context from adjacent patches (see Figure 3). To mitigate this, we allow crops to overlap so each patch has context from at least some neighboring patches. Patch features from the overlap are *not* passed to the connector or LLM so that the passed patch features exactly tile the high-resolution image. Overlapping slightly reduces the tiled image resolution, but this can be offset by using more crops. Overlapping significantly improves results, as shown in Section 6.

Vision-language connector. Once crops are encoded by the vision encoder, we build patch features by concatenating features from the third-to-last and tenth-from-last ViT layers, which improves performance slightly over using a single layer. Each 2×2 patch window is then pooled into a single vector using a multi-headed attention layer, where the mean of the patches serves as the query. This attention pooling outperforms simple feature concatenation (see Section 6). Finally, pooled features are mapped to the LLM’s embedding space via an MLP.

Arranging vision tokens. Pooled patch features (vision tokens) are sequenced left-to-right, top-to-bottom, starting with patches from the low-resolution full image, followed by high-resolution crop patches arranged in row-major or-

der. Special tokens are inserted to mark the start and end of both low- and high-resolution patch sequences, with row-end tokens added between rows to indicate row transitions.

Dropout. Residual dropout is applied to the LLM. During pre-training, dropout is applied only to text tokens to encourage reliance on the image rather than language priors. This is not used in fine-tuning, as shorter target responses result in too little dropout. Text-only dropout enhances captioning and downstream performance, as shown in Section 6.

Multi-annotated images. Our multimodal data often includes multiple annotations per image (*e.g.*, VQA v2.0 has multiple question-answer pairs). To train efficiently, we arrange *all* of text annotation tokens for an image in one long sequence, masking attention so tokens for each annotation attend to the image tokens, each other, but not to tokens from different annotations. This setup is equivalent to training on individual image-text pairs but avoids redundant image encoding, reducing the number of processed images by two-thirds and shortening training time by over half, with only a 25% increase in sequence length for our data mix.

3. Data

PixMo contains seven datasets, three with human annotations and four created with synthetic data generation pipelines (see Figure 1). Below, we describe these datasets and data collection methods; additional details and examples are in the Appendix.

PixMo-Cap. We collected PixMo-Cap as a source of high-quality pre-training data, featuring a diverse set of images paired with highly detailed dense captions. We began by sourcing web images across ~70 diverse topics (*e.g.*, street signs, memes, food, drawings, websites, blurry photos, *etc.*). For each image, three annotators initially provided detailed descriptions by speaking for at least 60 seconds. In later stages, we used one annotator per image with a 90-second minimum, which improved efficiency without sacrificing quality. We prompted the annotators with seven questions to answer, detailed in the Appendix.

The annotators’ audio was transcribed using a standard speech-to-text system, yielding raw transcripts. A final high-quality image caption was then created by prompting a language-only LLM to summarize multiple raw transcripts per image or, for single transcripts, to enhance its quality (*e.g.*, removing spoken artifacts, normalizing style). In total, we collected 712k distinct images with 1.3M transcripts and captions. Our captions average 196 words, compared to 11 words in COCO captions [17] and 37 words in localized narratives [93], highlighting their greater detail.

PixMo-AskModelAnything. We collected this data to enable the model to answer diverse questions it might encounter in real-world use. To create image-question-answer

triplets, annotators worked with a language-only LLM. An annotator selected an image from a large pool and wrote a question about it. Then, we ran a standard non-VLM OCR model and a PixMo-Cap-trained model on the image. The language-only LLM answered the question from the OCR data and dense caption. The annotator could accept or reject the answer and if rejected, they specified the issue and requested a revision until the answer was satisfactory. We collected 162k question-answer pairs in 73k images.

PixMo-Points. We collected pointing data to achieve three goals: (1) enable the model to point to items described by text, (2) enable the model to count by pointing, and (3) use pointing as a form of visual explanation when answering questions. For the first two goals, annotators were asked to point at something in an image, describe it, and then point to each instance of it in the image, ensuring exhaustive coverage. We also collected “not present” data so models can learn to handle cases where an item is *not* in the image. Pointing data also naturally supports answering counting questions with a chain-of-thought formed by the sequence of points. This resulted in 2.3M question-points pairs from 223k images. To enable points as explanations, we adapted the PixMo-AskModelAnything pipeline to let annotators pass the LLM a list of text-annotated points, prompting the LLM to use them in its answer when relevant. We collected 79k point-explanation annotations on 14k images.

PixMo-CapQA. We generated 214k question-answer pairs, covering diverse topics and styles, from 165k images by prompting a language-only LLM to ask and answer questions given only the ground-truth caption for an image.

PixMo-Docs. We used an extensive and carefully tuned prompting framework to prompt an LLM to generate code for 255k text and figure-heavy images, including charts, documents, tables, and diagrams. We then prompted the LLM to generate 2.3M question-answer pairs based on privileged access to the code (the images were not used).

PixMo-Clocks. We rendered synthetic clocks matched with a time-telling question-answer pair. The images use ~50 different watch bodies and ~160k realistic diverse watch faces set to random times. We collected 826k examples.

PixMo-Count. We used a standard non-VLM object detector [136] on web images to create image and counting QA pairs. For each image, we selected the class with the most detections after strict confidence thresholding. Following CountBenchQA [10], we manually verified 120 samples per count from 2 to 10, creating validation and test sets of 540 images each. These diverse images form a more challenging counting QA set than CountBenchQA, which has reported limitations [10]. The remaining samples with counts between 0 and 10 form a training set of 36k images, each annotated with points (object centers) and a QA pair.

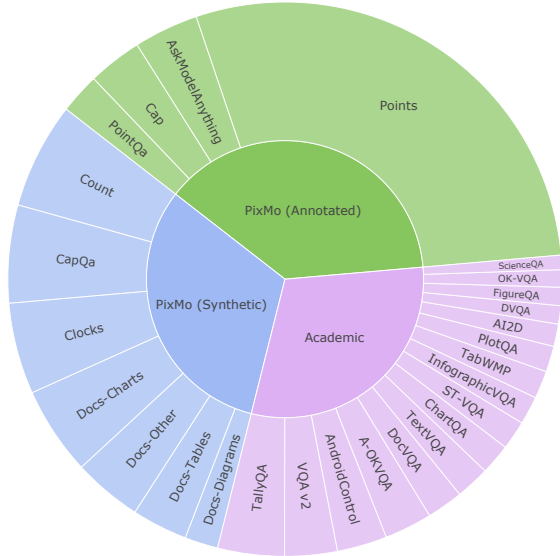


Figure 4. Datasets used for fine-tuning, shown in proportion to their sampling rates. Green denotes human-annotated data we collected, blue denotes synthetic data we generated, and purple represents pre-existing academic datasets. PixMo-Docs has been subdivided into charts, tables, diagrams, and other.

4. Training

Pre-training. We pre-train all parameters on PixMo-Cap to generate either the caption or one of the audio transcripts for a given image. A prompt specifies which one to generate and, 90% of the time, includes a length hint to guide the model’s output length. This hint improves caption and pre-training quality, as shown in Section 6.

Previous work has often included a separate training stage to tune only the vision-language connector [10, 27, 59, 69, 106]. We find this step unnecessary when pre-training on PixMo-Cap (see Section 6), also explored in [48]. Instead, we apply a higher learning rate with a shorter warmup for the connector parameters, allowing them to adjust more quickly at the start of training. Skipping this stage reduces training time and complexity, and eliminates the need for the noisy web-scale data typically used in this phase. We train for four epochs using AdamW [50, 73]. Full hyperparameters are in the Appendix.

Fine-tuning. We fine-tune the model on a mix of PixMo datasets and open-source training datasets, including: VQA v2.0 (COCO 2014 subset) [36], TextVQA [100], OK-VQA [81], ChartQA (re-weighted to balance human and augmented examples) [82], DocVQA [83], InfographicVQA [84], AI2D (transparent and opaque label boxes) [49], A-OKVQA [99], AndroidControl [62], ScienceQA [76], TabMWP [77], ST-VQA [11], TallyQA [2], DVQA [46], FigureQA [47], and PlotQA [86].

We sample datasets at rates proportional to the square root of their size, with manual down weighting of some

very large synthetic datasets (PlotQA, FigureQA, DVQA, and PixMo-Clocks). We observe that pointing tasks learn more slowly than QA tasks, so we significantly up-weight the pointing data. Final mixture rates are shown in Figure 4, with full details in the Appendix.

The academic datasets in this mixture teach specific skills and help the model perform well on corresponding benchmark test sets. However, these datasets often have answer styles that are not ideal for user interactions, as answers are usually very short and may reflect unique stylistic quirks from data collection (e.g., DocQA requires verbatim text from documents, while ChartQA specifies digits without commas). To prevent these styles from affecting user-facing responses, we prompt the model with a task-specific style tag (e.g., prefixing VQA v2.0 questions with “vqa2:”).

We use style tags for all datasets except PixMo-AskModelAnything, -CapQA, -Points, -Count and -Cap. For PixMo-Cap, we create ~30 prompts for caption generation. For pointing data, we create ~100 question templates that ask for the location or count of the target expression. The model then returns a list of points and, for counting questions, the total count. These prompts and templates are randomly sampled during training. We still use a style tag for pointing-as-an-explanation data since we find performance in this mode can be less reliable, so it should only be used when users request it.

The model outputs points as plain-text coordinates normalized between 0 and 100. When pointing to multiple items, points are ordered top-down, left-to-right, with each point numbered (see Figure 2 and details in the Appendix). Pointing enables a unique chain-of-thought approach to counting which improves performance (see Section 6).

5. Evaluation

We carefully evaluate on academic benchmarks because prompting, alignment with benchmark-specific answer styles, and use of benchmark training data can significantly affect performance. To complement this, we conduct a human evaluation to rank models based on user preference.

For academic benchmarking, we gather or compute results for all models on 10 common datasets and the PixMo-Count test set, which we include due to its higher difficulty compared to existing counting benchmarks. We prioritize author-published results but fill in missing results with the best previously reported values from technical reports or sources like the OpenVLM Leaderboard. If data is still missing, we compute it ourselves. Notably, computing results is challenging, as performance can vary significantly (e.g., by 10%) based on evaluation details. Additionally, critical information such as prompts or data processing steps is often unavailable, making it hard to reproduce results, highlighting the need for evaluation openness.

In our human evaluation, we collect 15k diverse image-

model	AI2D test [49]	ChartQA test [82]	VQA v2.0 testdev [36]	DocVQA test [83]	InfoQA test [84]	TextVQA val [100]	RealWorldQA [116]	MMMU val [129]	MathVista testmini [78]	CountBenchQA [10]	PixMo-Count test	Average	Elo score	Elo rank
API call only														
GPT-4V [88]	89.4	78.1	77.2	87.2	75.1	78.0	61.4	63.1	58.1	69.9	45.0	71.1	1041	10
GPT-4o-0513 [90]	94.2	85.7	78.7	92.8	79.2	77.4	75.4	69.1	63.8	87.9	59.6	78.5	1079	1
Gemini 1.5 Flash [103]	91.7	85.4	80.1	89.9	75.3	78.7	67.5	56.1	58.4	81.6	61.1	75.1	1054	7
Gemini 1.5 Pro [103]	94.4	87.2	80.2	93.1	81.0	78.7	70.4	62.2	63.9	85.8	64.3	78.3	1074	3
Claude-3 Haiku [7]	86.7	81.7	68.4	88.8	56.1	67.3	45.5	50.2	46.4	83.0	43.9	65.3	999	18
Claude-3 Opus [7]	88.1	80.8	66.3	89.3	55.6	67.5	49.8	59.4	50.5	83.6	43.3	66.7	971	21
Claude-3.5 Sonnet [7]	94.7	90.8	70.7	95.2	74.3	74.1	60.1	68.3	67.7	89.7	58.3	76.7	1069	4
Open weights only														
PaliGemma-mix-3B [10]	72.3	33.7	76.3	31.3	21.4	56.0	55.2	34.9	28.7	80.6	60.0	50.0	937	27
Phi3.5-Vision-4B [1]	78.1	81.8	75.7	69.3	36.6	72.0	53.6	43.0	43.9	64.6	38.3	59.7	982	19
Qwen2-VL-7B [111]	83.0	83.0	82.9	94.5	76.5	84.3	70.1	54.1	58.2	76.5	48.0	73.7	1025	14
Qwen2-VL-72B [111]	88.1	88.3	81.9	96.5	84.5	85.5	77.8	64.5	70.5	80.4	55.7	79.4	1037	12
InternVL2-8B [104]	83.8	83.3	76.7	91.6	74.8	77.4	64.2	51.2	58.3	57.8	43.9	69.4	953	23
InternVL2-Llama-3-76B [104]	87.6	88.4	85.6	94.1	82.0	84.4	72.7	58.2	65.5	74.7	54.6	77.1	1018	16
Pixtral-12B [3]	79.0	81.8	80.2	90.7	50.8	75.7	65.4	52.5	58.0	78.8	51.7	69.5	1016	17
Llama-3.2V-11B-Instruct [5]	91.1	83.4	75.2	88.4	63.6	79.7	64.1	50.7	51.5	73.1	47.4	69.8	1040	11
Llama-3.2V-90B-Instruct [5]	92.3	85.5	78.1	90.1	67.2	82.3	69.8	60.3	57.3	78.5	58.5	74.5	1063	5
Open weights + data († distilled)														
LLaVA-1.5-7B [69]	55.5	17.8	78.5	28.1	25.8	58.2	54.8	35.7	25.6	40.1	27.6	40.7	951	26
LLaVA-1.5-13B [69]	61.1	18.2	80.0	30.3	29.4	61.3	55.3	37.0	27.7	47.1	35.2	43.9	960	22
xGen-MM-interleave-4B† [119]	74.2	60.0	81.5	61.4	31.5	71.0	61.2	41.1	40.5	81.9	50.2	59.5	979	20
Cambrian-1-8B† [106]	73.0	73.3	81.2	77.8	41.6	71.7	64.2	42.7	49.0	76.4	46.6	63.4	952	25
Cambrian-1-34B† [106]	79.7	75.6	83.8	75.5	46.0	76.7	67.8	49.7	53.2	75.6	50.7	66.8	953	24
LLaVA OneVision-7B† [59]	81.4	80.0	84.0	87.5	68.8	78.3	66.3	48.8	63.2	78.8	54.4	72.0	1024	15
LLaVA OneVision-72B† [59]	85.6	83.7	85.2	91.3	74.9	80.5	71.9	56.8	67.5	84.3	60.7	76.6	1051	8
The Molmo family: Open weights, Open data, Open training code, Open evaluations														
MolmoE-1B	86.4	78.0	83.9	77.7	53.9	78.8	60.4	34.9	34.0	87.2	79.6	68.6	1032	13
Molmo-7B-O	90.7	80.4	85.3	90.8	70.0	80.4	67.5	39.3	44.5	89.0	83.3	74.6	1051	9
Molmo-7B-D	93.2	84.1	85.6	92.2	72.6	81.7	70.7	45.3	51.6	88.5	84.8	77.3	1056	6
Molmo-72B	96.3	87.3	86.5	93.5	81.9	83.1	75.2	54.1	58.6	91.2	85.2	81.2	1077	2

Table 1. We present academic benchmark results for 10 common datasets, plus a new counting benchmark, PixMo-Count, which features more challenging natural images than CountBenchQA. We categorize models into four groups: (top) proprietary models accessible only via API calls, (upper middle) models with released weights but closed data, (lower middle) models with released weights and training data (noting some of these use distillation (†) from proprietary VLMs via synthetic data), and (bottom) the Molmo family of models.

text prompt pairs and queried the VLMs for responses. We sample and present the resulting image-text-response triplets for all VLM pairings to a group of ~870 human annotators, who provide pairwise preference rankings. Across all model pairs, we gather over 325k ratings (~450 per model pair). From this data, we calculate an Elo ranking using the Bradley-Terry model, following [21].

For Molmo, we evaluate all academic datasets with 36 crops (up from 12 used in training), except for counting tasks, as pointing capabilities do not generalize well with different numbers of test crops. A small amount of high-res post-training can resolve this issue, see the Appendix.

When possible, we use relevant style prompts¹ (e.g., “vqa2:”). For evaluation-only datasets, we use the VQA v2.0 (for short answer) or A-OKVQA (for multiple choice)

¹We use AI2D with transparent boxes; see Appendix for opaque boxes.

style tags to elicit the often expected short answer style. For human evaluation, we omit style tags and use 12 crops, as some counting questions use pointing. Evaluators are only shown the output text, not the points.

Broadly speaking, the academic benchmark results and human evaluation agree, with the exception of Qwen2-VL [111], which performs strongly on the academic benchmarks and comparatively underperforms in the human evaluation. We highlight a few key results from Table 1:

- MolmoE-1B, our most efficient model based on the OLMoE-1B-7B mixture-of-experts LLM, nearly matches GPT-4V on academic benchmarks and Elo.
- Molmo models based on OLMo-7B-1024-preview and Qwen2 7B LLMs perform between GPT-4V and GPT-4o on academic benchmarks and Elo.
- Our best-in-class Qwen2 72B based model achieves the

ViT-L/14	cap F_1	11-avg
OpenAI CLIP 336px	54.1	76.9
MetaCLIP 336px	54.1	77.2
SigLIP-So400m 384px	54.4	77.1
DINOv2 336px	53.2	75.6

(a) **Vision encoder.** Encoders that were trained on noisy web-scale image-text pairs perform similarly (rows 1-3). Surprisingly, DINOv2, which is trained on images only (no text, no label supervision), is competitive on these tasks. MetaCLIP and DINOv2 are *fully* open.

cropping	cap F_1	11-avg
single	46.7	62.8
multi, no overlap	53.4	75.7
multi, overlap	54.1	76.9

(d) **Cropping.** Using the entire image only (single crop) performs poorly. Our novel overlapping crop method (see Figure 3), which prevents loss of context, performs the best.

# crops train, test	cap F_1	11-avg
4, 4	52.0	71.0
4, 12	52.0	74.1
4, 36	52.0	74.2
12, 12	54.1	74.9
12, 36	54.1	76.9
36, 36	54.0	77.2

(b) **Image resolution.** Using more crops at training and testing time generally improves performance. However, captioning and counting can perform poorly when # of crops are unequal, so for these tasks we always set the number of test crops equal to the training value.

setting	cap F_1	11-avg
off	53.0	76.2
on	54.1	76.9

(e) **Length conditioning.** Captioning with length hints is a superior pre-training task compared to captioning alone as evident by the improved captioning and downstream results.

pre-train, fine-tune	cap F_1	11-avg
off, off	53.1	74.6
off, on	53.1	76.6
on, on	53.7	77.0
on (text only), on	54.1	76.9

(c) **Dropout.** Dropout in the LLM improves pre-training and fine-tuning results. In pre-training, applying dropout to captioning text tokens only further improves results. This design may encourage the model to rely more on vision tokens rather than past text tokens.

2×2 pooling	cap F_1	11-avg
stacking	53.7	76.1
attention	54.1	76.9

(f) **Pooling.** Pooling 2×2 windows of vision tokens using mean-query attention performs better than simply stacking the four features as input to the vision-language connector MLP.

Table 2. **Model ablations.** Default settings are marked in gray. See the Appendix for additional ablations.

highest academic benchmark score and ranks second in Elo, just behind GPT-4o.

- Our best model also outperforms many state-of-the-art proprietary systems, including Gemini 1.5 Pro and Flash and Claude 3.5 Sonnet.

Molmo-72B also underwent an independent Elo evaluation via Chatbot Arena, where it outperforms all open models but ranks lower than several proprietary models (*e.g.*, GPT-4o and Claude 3.5 Sonnet).² The full results table is in the Appendix. The difference likely stems from the types of questions evaluated. While we cannot perform a full analysis since the questions are not public, we do note our data includes many counting and image-description questions which are particular strengths of Molmo.

Molmo excels at answering questions about natural images, matching or outperforming all models on the zero-shot RealWorldQA benchmark and achieving state-of-the-art results on the highly competitive VQA v2.0. On OCR-centric benchmarks (ChartQA, DocQA, InfoQA, TextVQA), Molmo surpasses other open models and some proprietary ones but trails slightly behind Qwen2-VL. On counting tasks (CountBenchQA and PixMo-Count), Molmo leads all models due to our new pointing data and chain-of-thought point-and-count abilities. However, on reasoning tasks (MMMU, MathVista) Molmo lags, likely because its training mix lacks data focused on advanced reasoning.

We conduct several additional skill-specific evaluations, summarized here with details in the Appendix. On a clock-reading benchmark [121], Molmo at all scales dramatically outperforms other VLMs including proprietary ones, but trails specialized non-VLM models [121]. To assess

Molmo’s potential for *action*, we tested Molmo-72B on AndroidControl [62], achieving 88.7% low-level and 69.0% high-level accuracy, comparable to the reported 83.2% and 70.8% in [62]. On NLP benchmarks, Molmo shows a slight performance drop versus its component LLM, which can be offset by additional text-only data. We also introduce a new pointing benchmark using SAM [51], where Molmo models at all scales demonstrate strong performance.

6. Ablations

We performed extensive ablations on model design (Table 2) and training data (Table 3), reporting two metrics: an F_1 metric (“cap F_1 ”) we developed to measure the precision and recall of captions generated by the model after pre-training (details in the Appendix), and the average accuracy on our 11 benchmark suite (“11-avg”), using validation sets when available. We report F_1 because we believe it reflects broad-range image understanding learned during pre-training, and many of our modeling design choices were based on this evaluation since it does not require running the more costly fine-tuning stage. While performing these ablation we observed captioning improvements generally, but not always, correspond to benchmark suite improvements. Our ablations test modifications to the Molmo-7B-D model configuration with key findings summarized below and further details in Table 2 and 3 captions and the Appendix.

Model ablations. We vary several design choices, finding:

- Vision encoders trained on noisy web-scale data (CLIP, SigLIP, MetaCLIP) all work roughly the same, including the *fully* open MetaCLIP (Table 2a).
- Excluding pointing and captioning, increasing the image resolution by increasing the number of crops generally

²<https://lmarena.ai/?leaderboard> vision arena, English category, accessed Nov. 13, 2024.

# PixMo-Cap images	cap F_1	11-avg
0 (0.0%)	-	74.9
89k (12.5%)	49.6	75.5
178k (25.0%)	51.6	76.3
356k (50.0%)	52.6	76.2
712k (100.0%)	54.1	76.9

data	cap F_1	11-avg
stage 0.5 LAION	53.9	76.9
ShareGPT4V+o (158k images)	36.3	74.9
<i>PixMo-Cap images:</i>		
our raw transcripts only	45.2	76.4
our cleaned transcripts only	53.0	76.5
our raw & cleaned transcripts captioned by GPT-4o	54.1	76.9
	52.9	77.5

data	11-avg
academic only	72.5
plus PixMo-Docs	74.0
PixMo- $\star$ plus academic	76.9
remove PixMo-AMA	76.8
remove PixMo-CapQA	77.0
remove PixMo-Docs	75.8
remove PixMo-Clocks	76.9
remove pointing task	76.2

(a) **PixMo-Cap scaling.** Increasing the quantity of PixMo-Cap captioning data in both pre-training and fine-tuning improves captioning and downstream tasks.

(b) **Pre-training data.** Our human annotated data performs on par with distilling GPT-4o by using it to caption the same set of images, demonstrating the effectiveness of our data.

(c) **Supervised fine-tuning data.** The PixMo- $\star$ datasets not only give the model new capabilities (e.g., pointing), but also generally improve results on the 11 dataset benchmark.

Table 3. **Data ablations.** Default settings are marked in gray .

strategy	CBQA	PCQA
count	87.9	80.2
point then count	89.4	86.3
count then point	81.5	77.6
pointing + regex	88.4	85.4

order	CBQA	PCQA
on	89.4	86.3
off	85.4	74.1

points, length	CBQA	PCQA
actual, correct	89.4	86.3
random, correct	85.9	76.3
random, random	76.3	75.7

tokens	CBQA	PCQA
plain-text	89.4	86.3
special	85.8	80.9

(a) **Counting strategy.** Pointing is the key ingredient in Molmo’s counting abilities.

(b) **Point order.** Training on ordered points (top-down, left-right) is better than unordered points.

(c) **Inference compute.** Simply increasing inference compute with extra tokens does not help.

(d) **Special point tokens.** Encoding point coordinates as plain-text works best.

Table 4. **Counting ablations.** Defaults are in gray . CBQA is the CountBenchQA test set and PCQA is the PixMo-Count validation set.

improves performance, and tuning at a high resolution yields a slight gain (Table 2b).

- Our novel text-only dropout used in pre-training improves captioning performance (Table 2c).
- Using multiple crops is critical, and our overlapping crop design yields significant improvements (Table 2d).
- Our novel length-conditioned captioning is a strong pre-training task, improving downstream results (Table 2e).
- Attention pooling yields improvements to both metrics over the baseline feature stacking approach (Table 2f).

Data ablations. We train with various data choices, finding:

- Scaling PixMo-Cap from 0 to 712k images significantly improves captioning and benchmark metrics (Table 3a).
- Adding noisy web-scale data does not help. PixMo-Cap does better than a similar amount of data from ShareGPT4v/o, and roughly as well as captions generated by GPT-4o on the PixMo-Cap images (Table 3b).
- The PixMo fine-tuning datasets improve benchmark task performance beyond the academic datasets, mainly by improving OCR and counting tasks (Table 3c).

Counting. We ablate several details of counting using models fine-tuned on just PixMo-Points and PixMo-Count data:

- Chain-of-thought point-then-count performs significantly better than generating a count only or generating a count followed by points. Pointing on its own, counted by regular expression, is only slightly worse (Table 4a).
- Training with points in a predictable spatial order (top-down, left-to-right) is important (Table 4b).
- Training with random locations and/or numbers of points (Table 4c) performs worse despite equalizing the amount of compute used during inference.
- Representing point coordinates in plain-text works bet-

model	Elo score	win % vs. default
<i>Claude-3.5 Sonnet</i>	1047	65%
PixMo-Cap w/ GPT-4o captions (Table 3b)	1018	55%
PixMo- $\star$, remove PixMo-CapQA (Table 3c)	1015	50%
Molmo-7B-D default	1014	n/a
PixMo- $\star$, no academic datasets (Table 3c)	1013	42%
<i>GPT-4V</i>	1010	47%
DINOv2 vision encoder (Table 2a)	999	45%
PixMo- $\star$, remove PixMo-AMA (Table 3c)	995	40%
no PixMo-Cap data (Table 3a)	990	35%
academic only (Table 3c)	897	17%

Table 5. **Elo** scores and win rates (excluding ties) for select ablations of Molmo-7B-D and two *API-only* models for context.

ter than introducing special location tokens (Table 4d).

Human evaluation. A human evaluation of select ablation models in Table 5 shows that PixMo data, especially PixMo-Cap and PixMo-AskModelAnything, is important for generating responses that users like. Academic datasets help, but perform poorly in isolation. GPT-4o captions on our images also perform well, likely due to recent advances in GPT and the diversity of our image collection (e.g., the ShareGPT datasets significantly underperform our data even at the same scale, see Tables 3b and 3a). While distilling from proprietary models might be effective, we believe that it is critical for the scientific community to understand how to train competitive VLMs without doing so.

7. Conclusion

We have presented Molmo, a state-of-the-art VLM with fully open training code and open, non-distilled, training data. Molmo is a significant step towards closing the gap between proprietary and open VLMs.

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